Predicting Chess Blunders Under Time Pressure

Group: SixSixGoSmooth

Group Information and Contributions

Wenxi GUAN (60237406): will collect, organize, and preprocess the data, and prepare the accompanying data documentation.

Canyang CHEN (60775208): will design the research methodology and plan the modelling workflow, and contribute to the relevant sections of the report.

Geyunqing SHI (61028298): will construct and implement models, and contribute to the methodology description and presentation materials.

Siying CHEN (61104552): will refine the modelling approach and integrate the overall work, and coordinate report writing and slide preparation.

Yibo LIU (60504525): will implement the experimental plan, build the experimental pipeline, and train and test the models.

Senyuan WU (60231487): will evaluate model performance, analyze and visualize the results, summaries the findings, and contribute to writing and presenting the experimental section.

Introduction

We propose to predict whether a player will make a serious error on the next move, using information available when their turn begins. Chess provides recorded decisions, player ratings and an engine-based measure of move quality. Studying how remaining time adds to rating and board information makes this a practical problem in probabilistic classification. The project connects regularization, nonlinear modelling, model assessment and interpretation from SDSC5001. Its value is to quantify when clock information improves prediction and whether a flexible model earns its extra complexity.

Research Questions

1. Does remaining time improve next-move blunder prediction beyond player rating and the current position?
2. Is the adjusted association nonlinear, and does it differ across player ratings or game phases?
3. Does gradient boosting improve probability accuracy and discrimination over an interpretable logistic model?

Dataset Description

We will use the September 2026 rated standard-chess PGN archive from the Lichess Open Database, the latest complete month available as of 27 October 2026. It contains 89,616,462 games in a 29.2 GB compressed file. A recent complete month provides a contemporary sampling frame without partial-month coverage. The CC0 archive contains ratings, timestamps and move sequences; clocks and engine evaluations are available for subsets of games. The analysis unit will be one recorded move, nested within a game and player.

Dataset Construction and Scope

We will retain standard games with a 180+0 time control, meaning three minutes per player without increment, numeric ratings, valid move sequences, clocks and engine annotations. Using one time control avoids mixing different clock regimes. A seeded reservoir sample will select up to 100 candidate games per UTC day, at most 3,000 across the month, without balancing on errors or outcomes. Quality checks may reduce this total. We anticipate roughly 100,000 to 200,000 usable moves and approximately 16 predictors before encoding; these are planning estimates. The final report will give actual counts, class prevalence and exclusions by day and rating.

We will scan the archive once using lightweight header and annotation filters, sample candidates by day, and fully parse only retained PGNs. Streaming avoids storing the decompressed archive, but still requires reading the 29.2 GB source. A 200-game pilot will measure coverage, processing time and class counts before freezing the sample. Invalid sampled games will be excluded without replacement. If too few qualify, we will simplify subgroup analysis. Existing evaluations avoid full-archive engine reanalysis. The engine-analysed 180+0 subset from this month defines our target population; findings need not generalize to other periods or time controls.

Investigation Plan

We will use Python with python-chess for board reconstruction, pandas for tabular processing, scikit-learn for modelling and matplotlib for figures. Generative AI assists proposal drafting and may assist debugging; the team will check code, references and interpretations, and all empirical results will come from reproducible scripts.

Prediction Time and Blunder Definition

A prediction is made at the start of the mover's turn, before any time is spent choosing the current move. A PGN clock annotation records time after a move . Therefore, the mover's starting clock comes from their previous move, two plies earlier; their first turn starts at 180 seconds. The opponent's clock comes from the immediately preceding ply. Historical thinking times use only completed earlier turns. We will exclude invalid clock histories and never use the current move's elapsed time, resulting board, annotation symbols or game result as predictors.

Let c_before and c_after be adjacent engine evaluations in centipawns, both converted from White's perspective to the current mover's perspective. Following the Lichess mapping , define a winning-chance proxy W(c) = 100 / (1 + exp(-0.00368208c)). Our primary label is y = 1 when $W(c_{before}) - W(c_{after})$ is at least 10 percentage points; otherwise y = 0. This is a study-defined serious-error threshold, not an official Lichess blunder label or a calibrated probability for every rating group. It reduces the distortion of a fixed pawn-loss threshold in already decisive positions. The main sample requires finite evaluations on both sides of a move; missing and mate evaluations will be counted and excluded, rather than replaced by arbitrary centipawn caps.

Predictors and Exploratory Analysis

Candidate inputs are both starting clocks, mover rating, rating difference, full-move number, colour, previous own thinking time and its three-turn mean, history-missing indicators, material advantage, total non-pawn material, legal-move count, check status, phase and the pre-move evaluation proxy. All board inputs use the position before the move. Opening means moves 1-10; thereafter, endgame means combined non-pawn material at most 14 points (N/B = 3, R = 5, Q = 9), with other positions classified as middlegame. These operational phases and legal-move counts are coarse descriptors. We will plot error rates and sample counts across clock bands, ratings and phases, and inspect missingness and class balance.

Models and Validation

The benchmark predicts the training-set error prevalence. Our main interpretable model is L2-regularized logistic regression, with spline terms for log(1 + remaining seconds), rating and move number, plus clock-by-rating and clock-by-phase interactions. A histogram gradient boosting classifier will test whether more flexible relationships improve prediction . We will compare identical feature sets and tune regularization or tree complexity with a small, pre-specified search of at most 12 configurations per model.

Games dated 24-30 September 2026 will form a locked final test set. Within 1-23 September, three expanding validation windows will train on days 1-8, 1-13 and 1-18, and validate on days 9-13, 14-18 and 19-23, respectively. Entire games remain together. Imputation, scaling, spline fitting and tuning occur inside each training window ; boosting's random internal early stopping will be disabled. We will select by mean validation log loss, refit on days 1-23 and evaluate the frozen models on the final week once.

Log loss is the primary metric because useful predictions must assign sensible probabilities. We will also report average precision with test prevalence, ROC AUC, Brier score and calibration plots. Any precision, recall or F1 operating threshold will be chosen using validation data. Training will retain the natural class distribution. Paired bootstrap resampling of whole test games will give 95% intervals for differences in log loss and average precision; repeated-player dependence will remain a limitation of these intervals.

Planned Experiments and Robustness Checks

Ablations will remove all clock and thinking-time inputs, and separately remove the pre-move engine evaluation, to measure their incremental value. We will repeat the label definition at 5- and 15-point losses and with a 150-centipawn loss, using the same temporal split. A mate-inclusive sensitivity analysis will map forced wins and losses to proxy endpoints of 100 and 0. A separate player-disjoint check will remove training games involving either player in the final test set and report the retained sample and performance.

For interpretation, we will plot adjusted logistic risk curves only over clock ranges supported within each rating and phase subgroup, alongside observed rates and counts. We will compare eligible analysed games with a small unfiltered 180+0 sample on rating, date and length to describe selection. Analysis availability, variable engine quality, unmeasured position difficulty and the tendency for clocks to run down as games progress prevent a causal interpretation of time pressure.

Expected Outcomes

We expect to deliver a documented move-level dataset, a reproducible modelling pipeline and an honest comparison of predictive accuracy, calibration and interpretability. We hypothesize that low remaining time adds predictive information, with nonlinear associations that may differ by skill and phase. These hypotheses may be unsupported, and subgroup directions will be treated as exploratory. The contribution will be measured predictive gains and qualified risk patterns, rather than a claim that time pressure causes errors or that the model generalizes to all chess players.

Work Schedule

By 31 October: finalize scope and submit the proposal. From 1-7 November: complete the pilot, freeze sampling and audit clocks and labels. From 8-14 November: finish exploratory analysis and tune models. From 15-21 November: run the frozen test and pre-specified robustness checks. From 22-28 November: complete the report, poster and code appendix, with shared review before submission.

References

Lichess Open Database. September 2026 rated standard-chess archive and documentation. https://database.lichess.org/